

GCP Data Engineer Interview Q&A (follow [Ahmed Mohiuddin | LinkedIn](#))

1. Briefly Introduce Yourself and Explain Your Most Recent Project in Detail

I am a cloud engineer with a strong focus on data engineering and analytics, specializing in GCP services like BigQuery, Dataflow, and Pub/Sub. My most recent project involved designing and implementing a **centralized data lake on GCP** for a healthcare provider. The solution consolidated claim-related data from various sources into Google Cloud Storage, processed it using Dataflow pipelines, and enabled real-time analytics in BigQuery. Key challenges included ensuring data security, regulatory compliance (e.g., HIPAA), and providing near real-time insights to optimize claim processing.

2. Why Did You Choose to Focus on GCP for Your Cloud Engineering Career?

GCP offers industry-leading data analytics and machine learning services, such as BigQuery and Vertex AI, which align with my passion for data-driven solutions. Its robust security features, user-friendly interfaces, and integration capabilities make it ideal for innovative and scalable cloud engineering projects.

3. Explain the Difference Between Partitioned and Clustered Tables in BigQuery. How to Decide Which One to Use?

- **Partitioned Tables:** Data is divided into segments based on a specified column (e.g., date), reducing scan time for queries that filter by that column.
- **Clustered Tables:** Data is organized by sorting rows based on one or more columns, improving performance for queries filtering by those columns.

When to Use:

- Use partitioning for large datasets primarily filtered by time-based fields.
 - Use clustering for queries on smaller datasets or those heavily filtered by non-date fields.
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4. Write a SQL Query in BigQuery to Find the Top 5 Most Purchased Products

```
SELECT
  product_id,
  COUNT(*) AS purchase_count
FROM
  `sales_data.transactions`
GROUP BY
  product_id
ORDER BY
```

`purchase_count DESC`

`LIMIT 5;`

5. How to Secure a Bucket in Google Cloud Storage to Ensure Only Specific Users Can Access It?

- **Remove Default Permissions:** Disable “AllUsers” and “AllAuthenticatedUsers” access.
 - **Apply IAM Roles:** Assign roles like **Storage Object Viewer** or **Storage Object Admin** only to specific users or service accounts.
 - **Bucket Policy Only:** Enable uniform bucket-level access for consistent permission control.
 - **Use Signed URLs:** Provide temporary access for external users if necessary.
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6. What Is Lifecycle Management in GCS, and How Would You Use It to Manage Storage Costs?

Lifecycle management automates object transitions between storage classes (e.g., Standard to Coldline) based on defined conditions like object age or access frequency.

- Example: Move files older than 90 days to **Nearline** and delete files after 365 days to reduce costs.
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7. Differences Between an Operator and a Sensor in Airflow

- **Operator:** Executes a task (e.g., running a script or query).
 - **Sensor:** Waits for a specific condition to be met (e.g., file availability).
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8. How to Handle Failures in Airflow Tasks?

- **Retries:** Specify the number of times a task should retry after failure.
 - **Backoff:** Define an interval that increases exponentially between retries to avoid overwhelming resources.
 - **Usage:** Use retries and backoff for temporary network issues or service unavailability.
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9. Design a DAG in Python That Extracts Data from an API Daily, Processes It, and Loads It Into BigQuery

```
from airflow import DAG
```

```
from airflow.providers.google.cloud.operators.bigquery import BigQueryInsertJobOperator
```

```
from airflow.operators.python import PythonOperator
```

```

from datetime import datetime

def extract_api_data(**kwargs):
    # Logic to extract data from API and save it locally or to Cloud Storage
    pass

def process_data(**kwargs):
    # Logic to transform the extracted data
    pass

default_args = {
    'start_date': datetime(2023, 1, 1),
    'retries': 3,
}

with DAG('api_to_bigquery', default_args=default_args, schedule_interval='@daily') as dag:
    extract = PythonOperator(task_id='extract_data', python_callable=extract_api_data)
    process = PythonOperator(task_id='process_data', python_callable=process_data)
    load_to_bq = BigQueryInsertJobOperator(
        task_id='load_to_bigquery',
        configuration={
            'query': {
                'query': 'LOAD DATA QUERY',
                'useLegacySql': False,
            },
        },
    )

    extract >> process >> load_to_bq

```

10. How Does Pub/Sub Ensure Message Delivery?

- **At-Most-Once Delivery:** Messages are delivered without retries, potentially leading to data loss.
 - **At-Least-Once Delivery:** Messages are retried until acknowledged by subscribers, ensuring reliable delivery but risking duplicates.
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11. What Are Dead-Letter Topics in Pub/Sub?

Dead-letter topics capture undeliverable messages that fail multiple delivery attempts. They help isolate problematic messages for debugging and prevent disruptions to the main topic.

12. Steps to Build a Dataflow Pipeline for a Batch Processing Task

1. Design the pipeline using **Apache Beam SDK**.
 2. Specify input data source (e.g., Cloud Storage).
 3. Apply transformations (e.g., filters, aggregations).
 4. Write output to a sink (e.g., BigQuery).
 5. Deploy the pipeline on **Dataflow Runner**.
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13. Difference Between Windowing and Triggering in Apache Beam

- **Windowing:** Defines how data is grouped (e.g., fixed or sliding time windows).
 - **Triggering:** Determines when results are emitted (e.g., after event completion or periodic intervals).
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14. SQL Query to Find Duplicate Records in a Table

```
SELECT
    column1, column2, COUNT(*) AS count
FROM
    `dataset.table`
GROUP BY
    column1, column2
HAVING
    count > 1;
```

15. Query to Calculate Running Total of Sales by Month

```
SELECT  
  
    DATE_TRUNC(date, MONTH) AS month,  
  
    SUM(sales) OVER (ORDER BY DATE_TRUNC(date, MONTH)) AS running_total  
  
FROM  
  
    `dataset.table` ;
```

16. Challenges in Migrating On-Prem Databases to Cloud SQL

- **Latency:** Mitigate with direct interconnect or VPN.
 - **Downtime:** Use replication for minimal impact.
 - **Schema Compatibility:** Address with tools like **Database Migration Service** or manual adjustments.
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17. How to Handle Schema Changes in an ETL Pipeline?

- Implement schema evolution support using tools like **Apache Avro** or **Delta Lake**.
 - Use schema validation to detect and resolve mismatches early.
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18. Differences Between ETL and ELT

- **ETL:** Extract, transform data in a staging area, then load it into the destination. Example: Data warehouses.
 - **ELT:** Extract, load raw data into the destination, and transform using the destination's capabilities. Example: BigQuery.
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19. How to Integrate Looker with BigQuery?

- Connect Looker to BigQuery using a service account key.
 - Configure datasets and define LookML models for analytics.
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20. Role of LookML in Looker and Creating a Custom Dimension

LookML defines data structure and relationships in Looker.

- **Custom Dimension:**
- dimension: custom_dimension {
- sql: \${TABLE_NAME}.column_name * 2 ;;
- label: "Custom Calculation"}